

Causal Graphs

Overview

- Draw directed acyclic graphs (DAGs) to represent causal relationships.
- Identify important types of nodes in causal graphs, including confounders, mediators, and colliders.
- Express common challenges in causal attribution using causal graphs.
- Test for conditional independence using d-separation criteria.

Readings

https://mixtape.scunning.com/03-directed_acyclical_graphs

Understanding Directed Acyclic Graphs (DAGs)

Introduction to DAG Notation

- a DAG is a visual map showing how different variables interact and cause one another
- nodes (letters)
 - represent random variables (e.g., D, Y, X)
- directed arrows ($\rightarrow$)
 - represent a causal relationship
 - the arrow always points forward in time from cause to effect
- acyclic
 - causality flows in one direction only
 - you cannot have paths that loop back into themselves
 - e.g., $A \rightarrow B \rightarrow A$ is impossible

How They Are Used

- to map theory
 - they force researchers to explicitly state what they believe causes what
- to show what is not happening
 - leaving out an arrow between two nodes means you assume there is absolutely no direct causal relationship between them
- to identify bias

	○ DAGs expose hidden pathways that can corrupt statistical data
Paths and the Backdoor Criterion	
The Direct Path ($D \rightarrow Y$)	• the actual causal effect that you want to measure
Backdoor Paths (Spurious Connections)	• a backdoor path is any non-causal path that connects D to Y by starting with an arrow pointing at D (e.g., $D \leftarrow X \rightarrow Y$) • these paths create spurious correlations (false associations)
How to Achieve the Backdoor Criterion	• to find the true causal effect, you must close all open backdoor paths • you close a backdoor path by conditioning on (controlling for) the non-collider variables along that path • confounder X ○ a variable that jointly causes both the treatment (D) and the outcome (Y) ○ controlling for it blocks the bias
The Phenomenon of Colliding (Collider Bias)	• a Collider is a variable along a path where two arrows collide head-to-head into it • <i>Example Path:</i> $D \rightarrow X \leftarrow Y$ • here, X is the collider because both D and Y point directly at it
The Golden Rule of Colliders	• an unconditioned collider naturally blocks a backdoor path ○ it acts like a closed gate • CRITICAL ERROR ○ if you control for (condition on) a collider, you open the gate ○ this introduces a new type of bias called Collider Bias (or Endogenous Selection Bias)
Real-World Examples of Collider Bias	the "Bad Control" in Salary Data: • if you look at gender discrimination (D) on earnings (Y), controlling for occupation (X) can introduce collider bias if discrimination affects what occupations women can enter ($D \rightarrow X$) and talent affects occupation and earnings ($Y \leftarrow X$)

	Sample Selection Bias (Police Use of Force) • if researchers only analyze data collected after a police stop occurs, they are conditioning on a collider (the stop) • if race affects the probability of being stopped, and civilian behavior also affects the stop, analyzing only the stopped individuals creates a false statistical correlation between race and police severity
Summary Cheat Sheet for Research Design	• to fix Confounding Bias ○ locate the variables causing both D and Y, and control for them to close the backdoor path • to avoid Collider Bias ○ locate variables where two causal paths collide, and leave them alone ○ do not control for them • data alone is not enough ○ you cannot fix these biases with "big data" ○ you must use a theoretical model (a DAG) to understand how the data was generated in the first place